

PRACTICE SHEET

1. What is the value of x that satisfies the equation $\cos^{-1} x = 2 \sin^{-1} x$?
 (a) $1/2$ (b) -1
 (c) 1 (d) $-1/2$
2. What is the value of $\sin^{-1} \left(\sin \frac{2\pi}{3} \right)$?
 (a) $-\pi/3$ (b) $2\pi/3$
 (c) $-2\pi/3$ (d) $\pi/3$
3. What are the values of (x, y) satisfying the simultaneous equations $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$ and $\cos^{-1} x - \cos^{-1} y = \frac{\pi}{3}$?
 (a) $(0, 1)$ (b) $(\frac{1}{2}, 1)$
 (c) $(1, \frac{1}{2})$ (d) $(\frac{\sqrt{3}}{2}, 1)$
4. What is $\sin [\cot^{-1} \{ \cos(\tan^{-1} x) \}]$ where $x > 0$, equal to?
 (a) $\sqrt{\frac{x^2+1}{x^2+2}}$ (b) $\sqrt{\frac{x^2+2}{x^2+1}}$
 (c) $\frac{(x^2+1)}{(x^2+2)}$ (d) $\frac{(x^2+2)}{(x^2+1)}$
5. If $\sin^{-1} x = \tan^{-1} y$, what is the value of $\frac{1}{x^2} - \frac{1}{y^2}$?
 (a) 1 (b) -1
 (c) 0 (d) 2
6. If $\cos^{-1} \left(\frac{1}{\sqrt{5}} \right) = \theta$, then what is the value of $\operatorname{cosec}^{-1}(\sqrt{5})$?
 (a) $\left(\frac{\pi}{2} \right) + \theta$ (b) $\left(\frac{\pi}{2} \right) - \theta$
 (c) $\frac{\pi}{2}$ (d) $-\theta$
7. What is the value of $\tan^{-1} \left(\frac{m}{n} \right) - \tan^{-1} \left(\frac{m-n}{m+n} \right)$?
 (a) π (b) $\pi/2$
 (c) $\pi/4$ (d) $\pi/3$
8. What is $\tan(\cos^{-1} x)$ equal to?
 (a) $\frac{\sqrt{1+x^2}}{x}$ (b) $\frac{x}{1+x^2}$
 (c) $\frac{\sqrt{1+x^2}}{x}$ (d) $\sqrt{1+x^2}$
9. If $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$, then what is the value of x ?
 (a) $x = -\frac{1}{2}$ (b) $x = 1$
 (c) $x = \frac{1}{2}$ (d) $x = \frac{\sqrt{3}}{2}$
10. If $\sin^{-1} x + \sin^{-1} y = \pi/2$ and $\cos^{-1} x - \cos^{-1} y = 0$, then values x and y are respectively
 (a) $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}, \frac{1}{2}$
 (c) $\frac{1}{2}, -\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$
11. The formula $\sin^{-1} \{ 2x(1-x^2) \} = 2 \sin^{-1} x$ is true for all values of x lying in the interval
 (a) $[-1, 1]$ (b) $[0, 1]$
 (c) $[-1, 0]$ (d) $[-1/\sqrt{2}, 1/\sqrt{2}]$
12. If $\sin^{-1} \left(\frac{2a}{1+a^2} \right) - \cos^{-1} \left(\frac{1-b^2}{1+b^2} \right) = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$, then what is the value of x ?
 (a) a/b (b) ab
 (c) b/a (d) $\frac{a-b}{1+ab}$
13. If $\sin^{-1} x + \cot^{-1} (1/2) = \pi/2$, then what is the value for x ?
 (a) 0 (b) $1/\sqrt{5}$
 (c) $2/\sqrt{5}$ (d) $\sqrt{3}/2$
14. What is the principle value of $\operatorname{cosec}^{-1}(-\sqrt{2})$?
 (a) $\pi/4$ (b) $\pi/2$
 (c) $-\pi/4$ (d) 0
15. If $\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$, then what is the value of x ?
 (a) 1 (b) 7
 (c) 13 (d) 17
16. **Statement I.** If $-1 \leq x < 0$, then $\cos(\sin^{-1} x) = -\sqrt{1-x^2}$
Statement II. If $-1 \leq x < 0$, then $\sin(\cos^{-1} x) = \sqrt{1-x^2}$
 Which one of the following is correct in respect of the above statements?
 (a) Both statements I and II are independently correct and Statement II is the correct explanation of Statement I
 (b) Both Statements I and II are independently correct but Statement II is not the correct explanation of Statement I
 (c) Statement I is correct but Statement II is false
 (d) Statement I is false but Statement II is correct I.
17. If $\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$, then what is the value of x ?
 (a) 1 (b) 7
 (c) 13 (d) 17

ANSWER KEY

1.	a	2.	d	3.	b	4.	a	5.	a	6.	b	7.	c	8.	a	9.	d	10.	d
11.	d	12.	d	13.	b	14.	c	15.	c	16.	d	17.	c						

Solutions

Sol. 1. (a)

Given that $\cos^{-1} x = 2 \sin^{-1} x$

$$\Rightarrow \pi/2 - \sin^{-1} x = 2 \sin^{-1} x$$

$$\Rightarrow \pi/2 = 3 \sin^{-1} x$$

$$\Rightarrow \sin^{-1} x = \pi/6$$

$$\text{So, } x = \sin \pi/6 = 1/2$$

Sol. 2. (d)

$\sin^{-1}$

$$\left(\sin \frac{2\pi}{3} \right) = \sin^{-1} \left[\sin \left(\pi - \frac{2\pi}{3} \right) \right] = \sin^{-1} \left(\sin \frac{\pi}{3} \right) = \frac{\pi}{3}$$

$$\left[\sin \text{ce, } \sin \frac{2\pi}{3} = \sin \left(\pi - \frac{\pi}{3} \right) \right]$$

Sol. 3. (b)

Given that

On solving equation. (i) and (ii), we get

$$\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$$

$$\text{and } \cos^{-1} x - \cos^{-1} y = \pi/3$$

$$\Rightarrow \pi/2 - \sin^{-1} x - \pi/2 + \sin^{-1} y = \pi/3$$

$$[\text{since } \cos^{-1} x + \sin^{-1} x = \pi/2]$$

$$\Rightarrow -\sin^{-1} x + \sin^{-1} y = \pi/3$$

On solving equation (i) and (ii), we get

$$2 \sin^{-1} y = \pi \text{ and } 2 \sin^{-1} x = \pi/3$$

$$\Rightarrow \sin^{-1} y = \pi/2 \text{ and } \sin^{-1} x = \pi/6$$

$$\text{Hence, } y = \sin \pi/2 \text{ and } \sin^{-1} x = \pi/6$$

$$\Rightarrow x = 1/2 \text{ and } y = 1$$

Sol. 4. (a)

Let $\alpha = \tan^{-1} x \Rightarrow \tan \alpha = x$

$$\text{Then } \cos \alpha = \frac{1}{\sqrt{1 + \tan^2 \alpha}} = \frac{1}{\sqrt{1 + x^2}}$$

$$\Rightarrow \cos(\tan^{-1} x) = \left\{ \frac{1}{\sqrt{1 + x^2}} \right\}$$

$$\text{so, } \cot^{-1} \{ \cos(\tan^{-1} x) \} = \cot^{-1} \left\{ \frac{1}{\sqrt{1 + x^2}} \right\}$$

$$\text{let } \cot^{-1} \left(\frac{1}{\sqrt{1 + x^2}} \right) = \beta$$

$$\Rightarrow \cot \beta = \frac{1}{\sqrt{1 + x^2}}$$

and $\sin \beta =$

$$\frac{1}{\sqrt{1 + \cot^2 \beta}} = \frac{\sqrt{1 + x^2}}{\sqrt{x^2 + 1 + 1}} = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$$

$$\Rightarrow \sin [\cot^{-1} \{ \cos(\tan^{-1} x) \}] = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$$

Sol. 5. (a)

Let, $\sin^{-1} x = \tan^{-1} y = \theta$

$$\Rightarrow x = \sin \theta \text{ and } y = \tan \theta$$

$$\frac{1}{x^2} = \frac{1}{\sin^2 \theta} = \cot^2 \theta$$

$$\text{and } \frac{1}{y^2} = \frac{1}{\sin^2 \theta} = \cot^2 \theta$$

$$\Rightarrow \frac{1}{x^2} - \frac{1}{y^2} = \cot^2 \theta - \cot^2 \theta = 0$$

Sol. 6. (b)

$$\text{Let, } \cos^{-1} \left(\frac{1}{\sqrt{5}} \right) = \theta \Rightarrow \cos \theta = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \sec \theta = \sqrt{5} \Rightarrow \sec^{-1}(\sqrt{5}) = \theta$$

$$\Rightarrow \pi/2 - \csc^{-1}(\sqrt{5}) = \theta$$

$$(\because \sec^{-1} x + \csc^{-1} x = \pi/2)$$

$$\Rightarrow \csc^{-1}(\sqrt{5}) = \pi/2 - \theta$$

Sol. 7. (c)

The given expression is:

$$\tan^{-1} \left(\frac{m}{n} \right) - \tan^{-1} \left(\frac{m-n}{m+n} \right)$$

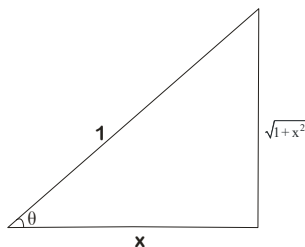
$$= \tan^{-1} \left(\frac{m}{n} \right) - \tan^{-1} \left(\frac{1 - \frac{n}{m}}{1 + \frac{n}{m}} \right)$$

$$= \tan^{-1} \left(\frac{m}{n} \right) - \tan^{-1} (1) + \tan^{-1} \left(\frac{n}{m} \right)$$

$$= \tan^{-1} \left(\frac{m}{n} \right) + \cot^{-1} \left(\frac{m}{n} \right) - \frac{\pi}{4} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Sol. 8. (a)

Let $\cos^{-1} x = \theta$



$$\Rightarrow \cos \theta = x = \sin \theta = \sqrt{1 - x^2}$$

$$\Rightarrow \tan \theta = \frac{\sqrt{1 - x^2}}{x} \text{ and } \theta = \cos^{-1} x$$

This can be represented by a triangle with hypotenuse = 1 and sides x and $\sqrt{1 - x^2}$

$$\Rightarrow \tan(\cos^{-1} x) = \frac{\sqrt{1 - x^2}}{x}$$

Sol. 9. (d)

As given

$$\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6} \quad \dots\dots(1)$$

And we know that

$$d \sin^{-1} x - \cos^{-1} x = \frac{\pi}{2} \quad \dots\dots(2)$$

On adding Eqs. (1) and (2) we get

$$2 \sin^{-1} x = \pi/6 + \pi/2 = 2\pi/3$$

$$\Rightarrow \sin^{-1} x = \frac{\pi}{3} \Rightarrow x = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

Sol. 10. (d)

Given, $\sin^{-1} x + \sin^{-1} y = \pi/2$

$$\text{and } \cos^{-1} x - \cos^{-1} y = 0$$

$$\Rightarrow \left(\frac{\pi}{2} - \sin^{-1} x \right) - \left(\frac{\pi}{2} - \sin^{-1} y \right) = 0$$

$$\Rightarrow \sin^{-1} y - \sin^{-1} x = 0$$

$$\Rightarrow \sin^{-1} y = \sin^{-1} x$$

From equations (i) and (ii), we get

$$2 \sin^{-1} x = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow x = \frac{1}{\sqrt{2}}$$

From equation (ii)

$$y = \frac{1}{\sqrt{2}}$$

Sol. 11. (d)

$\sin^{-1} \{ 2x(1-x^2) \} = 2 \sin^{-1} x$ is true

$$\forall x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

Sol. 12. (d)

Given,

$$\sin^{-1} \left(\frac{2}{1+a^2} \right) - \cos^{-1} \left(\frac{1+b^2}{1+b^2} \right) = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

$$\therefore 2 \tan^{-1} a - 2 \tan^{-1} b = 2 \tan^{-1} x$$

$$\Rightarrow \tan^{-1} a - \tan^{-1} b = \tan^{-1} x$$

$$\Rightarrow x = \frac{a-b}{1+ab}$$

Sol. 13. (b)

$$\text{Let } \sin^{-1} x + \cot^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{2}$$

$$\text{As we know } \cot^{-1} x = \sin^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$$

$$\therefore \sin^{-1} x + \cot^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x + \sin^{-1} \left(\frac{1}{\sqrt{1+\frac{1}{4}}} \right) = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} x + \sin^{-1} \left(\frac{2}{\sqrt{5}} \right) = \frac{\pi}{2} \quad \dots (1)$$

$$\text{Now, } \sin^{-1} x = \cos^{-1} \sqrt{1-x^2}$$

$$\therefore \sin^{-1} \left(\frac{2}{\sqrt{5}} \right) = \cos^{-1} \sqrt{1-\frac{4}{5}} = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$$

$\therefore$ from equation (1), we have

$$\sin^{-1} x + \cos^{-1} \left(\frac{1}{\sqrt{5}} \right) = \frac{\pi}{2}$$

$$\text{Since, } \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$\therefore x = \frac{1}{\sqrt{5}}$$

Sol. 14. (c)

Let the principal value of $\operatorname{cosec}^{-1}(-\sqrt{2}) = \theta$

$$\Rightarrow -\sqrt{2} = \operatorname{cosec} \theta \Rightarrow -\sqrt{2} = \frac{1}{\sin \theta}$$

$$\Rightarrow \sin \theta = -\frac{1}{\sqrt{2}} \Rightarrow \theta = -\frac{\pi}{4}$$

$$\text{Principal value of } \operatorname{cosec}^{-1}(-\sqrt{2}) = -\frac{\pi}{4}$$

Sol. 15. (c)

$$\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} \frac{5}{x} + \cos^{-1} \frac{\sqrt{x^2-144}}{x} = \frac{\pi}{2}$$

$$\left(\because \sin^{-1} x = \cos^{-1} \sqrt{1-x^2} \right)$$

But we know $\sin^{-1} y + \cos^{-1} y = \pi/2$

$$\therefore \frac{5}{x} = \frac{\sqrt{x^2-144}}{x}$$

$$\Rightarrow 5 = \sqrt{x^2-144}$$

$$\Rightarrow 25 = x^2 - 144 \Rightarrow x^2 = 169$$

$$\Rightarrow x = 13$$

Sol. 16. (d)

$$\text{I. } \cos(\sin^{-1} x) = \cos(\cos^{-1} \sqrt{1-x^2}),$$

$$= \sqrt{1-x^2} \quad \text{when } x \in [-1, 1]$$

$$\text{II. } \sin(\cos^{-1} x) = \sin(\sin^{-1} \sqrt{1-x^2}) = \sqrt{1-x^2}$$

when $x \in [-1, 1]$

Hence, Statement I is false and Statement II is correct.

Sol. 17. (c)

$$\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} \frac{5}{x} + \cos^{-1} \frac{\sqrt{x^2-144}}{x} = \frac{\pi}{2}$$

$$[\because \sin^{-1} A = \cos^{-1} \sqrt{1-A^2}]$$

$$\sin^{-1} A + \cos^{-1} A = \frac{\pi}{2}$$

$$\therefore \frac{5}{x} = \frac{\sqrt{x^2-144}}{x}$$

$$\therefore x = 13$$

NDA PYQ

1. If $\sin\left(\sin^{-1}\frac{1}{5} + \cos^{-1}x\right) = 1$, then what is the value of x ?
 (a) 0 (b) 1
 (c) $4/5$ (d) $1/5$
[NDA (I) 2011]
 2. Consider the following statements
 I. $\cos ec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = -\frac{\pi}{3}$
 II. $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$
 Which of the above statement(s) is/ are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) 2011]
 3. What is the principle value of $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$
 (a) $\pi/2$ (b) $\pi/3$
 (c) $\pi/4$ (d) $\pi/6$
[NDA (II) 2011]
 4. If $\tan^{-1}2$ and $\tan^{-1}3$ are two angles of a triangle, then what is the third angle?
 (a) $\tan^{-1}2$ (b) $\tan^{-1}4$
 (c) $\pi/4$ (d) $\pi/3$
[NDA (II) 2011]
 5. What is the value of $\cos\left\{\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13}\right\}$
 (a) $63/65$ (b) $33/65$
 (c) $22/65$ (d) $11/65$
[NDA (I) 2012]
 6. If $\tan^{-1}2$ and $\tan^{-1}3$ are two angles of a triangle, then what is the third angle?
 (a) $\tan^{-1}2$ (b) $\tan^{-1}4$
 (c) $\pi/4$ (d) $\pi/3$
[NDA (I) 2012]
 7. What is the value of $\sec^2 \tan^{-1}\left(\frac{5}{11}\right)$?
 (a) $121/96$ (b) $217/921$
 (c) $146/121$ (d) $267/121$
[NDA (I) 2012]
 8. If $\sin^{-1}1 = \sin^{-1}4/5 + \sin^{-1}x$, then what is x equal to?
 (a) $3/5$ (b) $4/5$
 (c) 1 (d) 0
[NDA (I)-2012]
 9. $\sin^{-1}\left(\frac{2a}{1+a^2}\right) + \sin^{-1}\left(\frac{2b}{1+b^2}\right) = 2\tan^{-1}x$ then x is equal to?
 (a) $\frac{a-b}{1+ab}$ (b) $\frac{a-b}{1-ab}$
 (c) $\frac{2ab}{a+b}$ (d) $\frac{a+b}{1-ab}$
[NDA (I) 2013]
 10. What is $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$ equal to?
 (a) $\pi/2$ (b) $\pi/3$
 (c) $\pi/4$ (d) $\pi/6$
[NDA (II) 2013]
 11. If x and y are positive and $xy > 1$, then what is $\tan^{-1}x + \tan^{-1}y$ to?
 (a) $\tan^{-1}\left(\frac{x+y}{1-xy}\right)$ (b) $\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$
 (c) $\pi - \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ (d) $\tan^{-1}\left(\frac{x-y}{1-xy}\right)$
[NDA (I) 2014]
 12. Consider the following statements
 I. $\tan^{-1}1 + \tan^{-1}(0.5) = \pi/2$
 II. $\sin^{-1}(1/3) + \cos^{-1}(1/3) = \frac{\pi}{2}$
 Which of the above statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (I) 2014]
 13. What is $\sin^{-1}\sin\left(\frac{3\pi}{5}\right)$ equal to?
 (a) $\frac{3\pi}{5}$ (b) $\frac{2\pi}{5}$
 (c) $\frac{\pi}{5}$ (d) None
[NDA (I) 2014]
 14. What is $\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{3}{5}$ equal to?
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
[NDA (II) 2014]
 15. The equation $\tan^{-1}(1+x) + \tan^{-1}(1-x) = \frac{\pi}{2}$ is satisfied by
 (a) $x = 1$ (b) $x = -1$
 (c) $x = 0$ (d) $x = 1/2$
[NDA (I) 2015]
- Directions (for next three) Consider**
 $x = 4\tan^{-1}\left(\frac{1}{5}\right)$, $y = \tan^{-1}\left(\frac{1}{70}\right)$
 $z = \tan^{-1}\left(\frac{1}{99}\right)$
16. What is x equal to?
 (a) $\tan^{-1}\left(\frac{60}{119}\right)$ (b) $\tan^{-1}\left(\frac{120}{119}\right)$
 (c) $\tan^{-1}\left(\frac{90}{169}\right)$ (d) $\tan^{-1}\left(\frac{170}{169}\right)$
[NDA (I) 2015]
 17. What is $x - y$ equal to?

(a) $\tan^{-1}\left(\frac{828}{845}\right)$ (b) $\tan^{-1}\left(\frac{8287}{8450}\right)$

(c) $\tan^{-1}\left(\frac{8281}{8450}\right)$ (d) $\tan^{-1}\left(\frac{8287}{8471}\right)$

[NDA (I) 2015]

18. What is the value of $x - y + z$ equal to?

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{4}$

[NDA (I) 2015]

19. The value of $\tan\left(2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right)$ is

(a) $-7/17$ (b) $5/16$

(c) $5/4$ (d) $7/17$

[NDA (II) 2015]

20. Consider the following statements

1. $\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{3}{5} = \frac{\pi}{2}$

2. $\tan^{-1}\sqrt{3} + \tan^{-1}1 = -\tan^{-1}(2 + \sqrt{3})$

Which of the above statements (s) is/are correct?

(a) Only 1 (b) Only 2

(c) Both 1 and 2 (d) Neither 1 or 2

[NDA (II) 2015]

21. Consider the following statement

1. $\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right) = \pi$

2. Their exist, $x, y \in [-1, 1]$, where $x \neq y$ such that

$\sin^{-1}x + \cos^{-1}y = \frac{x}{y}$

Which of the above statements is/are correct?

(a) Only 1 (b) Only 2

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) 2016]

22. Consider the following statements

1. There exists $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which $\tan^{-1}(\tan \theta) \neq \theta$.

2. $\sin^{-1}\left(\frac{1}{3}\right) - \sin^{-1}\left(\frac{1}{5}\right) = \sin^{-1}\left(\frac{2\sqrt{2(\sqrt{3}-1)}}{15}\right)$

Which of the above statements is/are correct?

(a) Only 1 (b) Only 2

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) 2016]

23. What is the value of $\cos(2\cos^{-1}(0.8))$?

(a) 0.81 (b) 0.56

(c) 0.48 (d) 0.28

[NDA (II) 2016]

24. Let x, y, z be positive real numbers such that x, y, z are in GP and $\tan^{-1}x, \tan^{-1}y$ and $\tan^{-1}z$ are in AP. Then which one of the following is correct?

(a) $x = y = z$ (b) $xz = 1$

(c) $x \neq y$ and $y = z$ (d) $x = y$ and $y \neq z$

[NDA-2017(I)]

25. The value of $\sin^{-1}\frac{3}{5} + \tan^{-1}\frac{1}{7}$ =

(a) 0 (b) 45°

(c) 60° (d) 90°

[NDA (II) 2017]

26. The principle value of $\sin^{-1}x$ lies in the interval

(a) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (b) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(c) $\left[0, \frac{\pi}{2}\right]$ (d) $[0, \pi]$

[NDA (II) 2017]

27. What is the principle value of $\sin^{-1}\sin\left(\frac{2\pi}{3}\right)$

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) $\frac{\pi}{3}$ (d) $\frac{2\pi}{3}$

[NDA (I) 2018]

28. What is $\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{3}{5}$ =

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

[NDA (I) 2018]

29. Consider the following values of x :

1. 8

2. -4

3. $1/6$

4. $-1/4$

Which of the above values of x are the solutions of equation $\tan^{-1}(2x) + \tan^{-1}(3x) = \pi/4$

(a) 3 only (b) 2 and 3 only

(c) 1 and 4 only (d) 4 only

[NDA (II) 2018]

30. What is the value of $\sin^{-1}\frac{4}{5} + \sec^{-1}\frac{5}{4} - \frac{\pi}{2}$

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$

(c) π (d) 0

[NDA (I) 2019]

31. $\sin^{-1}\left(\frac{2p}{1+p^2}\right) - \cos^{-1}\left(\frac{1-q^2}{1+q^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ then x is equal to

(a) $\frac{p+q}{1+pq}$ (b) $\frac{p-q}{1+pq}$

(c) $\frac{pq}{1+pq}$ (d) $\frac{p+q}{1-pq}$

[NDA (I) 2019]

32. What is $\tan\left\{2\tan^{-1}\frac{1}{3}\right\}$ =

(a) $2/3$ (b) $3/4$

(c) $3/8$ (d) $1/9$

[NDA (II) 2019]

33. The equation $\sin^{-1}x - \cos^{-1}x = \frac{\pi}{6}$ has

(a) no solution

(b) unique solution

(c) two solutions

(d) infinite number of solution

[NDA (I) 2021]

34. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ holds, when:

- (a) $x \in \mathbb{R}$
 (b) $x \in \mathbb{R} - \{-1, 1\}$ only
 (c) $x \in \mathbb{R} - \{0\}$ only
 (d) $x \in \mathbb{R} - [-1, 1]$ only

[NDA (I) 2021]

35. What is the value of the following:

$$\cot \left[\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right]$$

- (a) 6/17 (b) 7/16
 (c) 16/7 (d) 17/6

[NDA (II) 2021]

36. Let $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$ for $0 \leq x, y, z \leq 1$.

What is the value of $x^{1000} + y^{1001} + z^{1002}$?

- (a) 0 (b) 1
 (c) 3 (d) 6

[NDA (II) 2021]

37. If $\tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{x}{3} \right) = \frac{\pi}{4}$ where $0 < x < 6$, then what is x equal to?

- (a) 1 (b) 2
 (c) 3 (d) 5

[NDA (I) 2022]

38. If $3\sin^{-1} x + \cos^{-1} x = \pi$, then what is x equal to?

- (a) 0 (b) $\frac{1}{2}$
 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{3}}$

[NDA (I) 2022]

39. What is $\tan^{-1} \cot (\operatorname{cosec}^{-1} 2)$ equal to?

- (a) $\frac{\pi}{8}$ (b) $\frac{\pi}{6}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

[NDA 2022 (II)]

40. What is $2 \cot \left(\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3} \right)$ equal to?

- (a) -1 (b) 1
 (c) $3 + \sqrt{5}$ (d) $3 - \sqrt{5}$

[NDA-2023 (2)]

41. If $\sec^{-1} p - \operatorname{cosec}^{-1} q = 0$, where $p > 0, q > 0$; then what is the value of $p^{-2} + q^{-2}$?

- (a) 1 (b) 2
 (c) 1/2 (d) $\frac{1}{2\sqrt{2}}$

[NDA-2023 (2)]

42. What is $1 + \sin^2 \left(\cos^{-1} \left(\frac{3}{\sqrt{17}} \right) \right)$ equal to?

- (a) 25/17 (b) 8/17
 (c) 9/17 (d) 47/17

[NDA-2023 (2)]

43. What is $\sqrt{15 + \cot^2 \left(\frac{\pi}{4} - 2 \cot^{-1} 3 \right)}$ equal to?

- (a) 1 (b) 7
 (c) 8 (d) 16

[NDA-2024 (1)]

44. What is $\tan^{-1} \left(\frac{a}{b} \right) - \tan^{-1} \left(\frac{a-b}{a+b} \right)$

- (a) $-\frac{\pi}{4}$ (b) $\frac{\pi}{4}$
 (c) $\tan^{-1} \left(\frac{a^2 - b^2}{a^2 + b^2} \right)$ (d) $\tan^{-1} \left(\frac{2ab}{a^2 + b^2} \right)$

[NDA-2024 (1)]

45. If $\cos^{-1} x = \sin^{-1} x$, then which one of the following is correct?

- (a) $x = 1$ (b) $x = \frac{1}{2}$
 (c) $x = \frac{1}{\sqrt{2}}$ (d) $x = \frac{1}{\sqrt{3}}$

[NDA-2024 (1)]

46. If $4\sin^{-1} x + \cos^{-1} x = \pi$, then what is the value of $\sin^{-1} x + 4\cos^{-1} x$ equal to?

- (a) $\pi/2$ (b) π
 (c) $3\pi/2$ (d) 2π

[NDA-2024 (2)]

47. What is $\cot^2 (\sec^{-1} 2) + \tan^2 (\operatorname{cosec}^{-1} 3)$ equal to?

- (a) 11/12 (b) 11/24
 (c) 7/24 (d) 1/24

[NDA-2024 (2)]

48. If k is a root of $x^2 - 4x + 1 = 0$, then what is $\tan^{-1} k + \tan^{-1} \left(\frac{1}{k} \right)$ equal to?

- (a) $-\pi/2$ (b) 0
 (c) $\pi/4$ (d) $\pi/2$

[NDA-2025 (1)]

49. If $\tan^{-1} k + \tan^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{4}$, then what is the value of k?

- (a) 1 (b) $\frac{1}{2}$
 (c) 1/3 (d) $\frac{1}{4}$

[NDA-2025 (1)]

50. What is $\cot^{-1} 9 + \operatorname{cosec}^{-1} \left(\frac{\sqrt{41}}{4} \right)$, equal to?

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{2}$ (d) π

[NDA-2025 (2)]

51. What is the value of $\tan \left[\frac{1}{2} \sec^{-1} \left(\frac{2}{\sqrt{3}} \right) \right]$

- (a) $2 - \sqrt{3}$ (b) $2 + \sqrt{3}$
 (c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

[NDA-2025 (2)]

ANSWER KEY

1.	d	2.	c	3.	d	4.	c	5.	b	6.	c	7.	c	8.	a	9.	d	10.	c
11.	b	12.	b	13.	b	14.	a	15.	c	16.	b	17.	c	18.	d	19.	a	20.	a
21.	d	22.	b	23.	d	24.	a	25.	b	26.	b	27.	c	28.	b	29.	a	30.	d
31.	b	32.	b	33.	b	34.	a	35.	a	36.	c	37.	a	38.	c	39.	d	40.	c
41.	a	42.	a	43.	c	44.	b	45.	c	46.	c	47.	b	48.	d	49.	c	50.	a
51.	a																		

Solutions

Sol. 1. (d)

$$\sin \left[\sin^{-1} \left(\frac{1}{5} \right) + \cos^{-1} x \right] = 1$$

$$\Rightarrow \sin \left[\sin^{-1} \left(\frac{1}{5} \right) + \cos^{-1}(x) \right] = \sin \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} \frac{1}{5} + \cos^{-1} x = \frac{\pi}{2}$$

$$\left[\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right]$$

$$x = \frac{1}{5}$$

Sol. 2. (c)

$$\cos ec^{-1} \left(-\frac{2}{\sqrt{3}} \right) = -\frac{\pi}{3}$$

$$\sec^{-1} \left(\frac{2}{\sqrt{3}} \right) = \frac{\pi}{6}$$

Both statements are correct

Sol. 3. (d)

$$\sec^{-1} \left(\frac{2}{\sqrt{3}} \right) = \frac{\pi}{6}$$

Sol. 4. (c)

Given that $\tan^{-1} 2$ and $\tan^{-1} 3$ two angles of a triangles.

Let α be the third angle of the triangle, then
In $\triangle ABC$,

$$\tan^{-1} 2 + \tan^{-1} 3 + \alpha = \pi$$

$$\pi + \tan^{-1} \left(\frac{2+3}{1-(2)(3)} \right) + \alpha = \pi$$

$$\tan^{-1}(-1) + \alpha = 0$$

$$-\tan^{-1}(1) + \alpha = 0$$

$$\alpha = \tan^{-1}(1) = \frac{\pi}{4}$$

Sol. 5. (b)

$$\cos \left\{ \cos^{-1} \frac{4}{5} + \cos^{-1} \frac{12}{13} \right\}$$

$$= \frac{48}{65} - \sqrt{1 - \frac{16}{25}} \cdot \sqrt{1 - \frac{144}{169}}$$

$$= \frac{48}{65} - \sqrt{\frac{9}{25}} \cdot \sqrt{\frac{25}{169}} = 33/65$$

Sol. 6. (c)

Given that $\tan^{-1} 2$ and $\tan^{-1} 3$ two angles of a triangles.

Let α be the third angle of the triangle, then
In $\triangle ABC$,

$$\tan^{-1} 2 + \tan^{-1} 3 + \alpha = \pi$$

$$\pi + \tan^{-1} \left(\frac{2+3}{1-(2)(3)} \right) + \alpha = \pi$$

$$\tan^{-1}(-1) + \alpha = 0$$

$$-\tan^{-1}(1) + \alpha = 0$$

$$\alpha = \tan^{-1}(1) = \frac{\pi}{4}$$

Sol. 7. (c)

$$\sec^2 \tan^{-1} \left(\frac{5}{11} \right)$$

$$= 1 + \tan^2 \tan^{-1} \left(\frac{5}{11} \right) \left[\because \sec^2 \theta = 1 + \tan^2 \theta \right]$$

$$= 1 + \left\{ \tan \tan^{-1} \left(\frac{5}{11} \right) \right\}^2$$

$$= 1 + \left(\frac{5}{11} \right)^2 = 1 + \frac{25}{121} = \frac{146}{121}$$

Sol. 8. (a)

$$\sin^{-1} 1 = \sin^{-1} \left(\frac{4}{5} \right) + \sin^{-1} x$$

$$\frac{\pi}{2} = \sin^{-1} \left(\frac{4}{5} \right) + \sin^{-1} x$$

$$\frac{\pi}{2} = \cos^{-1} \left(\frac{3}{5} \right) + \sin^{-1} x$$

$$x = \left(\frac{3}{5} \right)$$

Sol. 9. (d)

$$\sin^{-1} \left(\frac{2a}{1+a^2} \right) + \sin^{-1} \left(\frac{2b}{1+b^2} \right) = 2 \tan^{-1} x$$

$$\Rightarrow 2 \tan^{-1} a + 2 \tan^{-1} b = 2 \tan^{-1} x$$

$$\left[\because \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1+x^2} \right) \right]$$

$$\Rightarrow (\tan^{-1} a + \tan^{-1} b) = \tan^{-1} x$$

$$\Rightarrow \tan^{-1} \left(\frac{a+b}{1-ab} \right) = \tan^{-1} x$$

$$\left[\because \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \right]$$

$$\therefore x = \frac{a+b}{1-ab},$$

Sol. 10. (c)

$$\left[\because \tan^{-1} A + \tan^{-1} B = \tan^{-1} \left(\frac{A+B}{1-AB} \right) \right]$$

$$\tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{1}{3} \right)$$

$$\tan^{-1} \left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} \right)$$

$$= \tan^{-1} \left(\frac{\frac{5}{6}}{\frac{5}{6}} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Sol. 11. (b)

$$\pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right)$$

Sol. 12. (b)

$$\Rightarrow \tan^{-1} 1 + \tan^{-1}(0.5)$$

$$\Rightarrow \tan^{-1} 1 + \tan^{-1} \left(\frac{1}{2} \right) = \tan^{-1} 1 + \cot^{-1} \left(\frac{1}{1/2} \right)$$

$$= \tan^{-1} + \cot^{-1} 2 \neq \pi/2$$

$$\left[\because \tan^{-1} x + \cot^{-1} x = \pi/2, \forall x \in \mathbb{R} \right]$$

Statement 1 is wrong

$$\sin^{-1}\left(\frac{1}{3}\right) + \cos^{-1}\left(\frac{1}{3}\right) = \pi/2$$

$$\left[\because \sin^{-1} A + \cos^{-1} A = \pi/2, \forall A \in R \right]$$

Statement 2 is correct

Sol. 13. (b)

We have,

$$\begin{aligned} \sin^{-1} \sin\left(\frac{3\pi}{5}\right) &= \sin^{-1} \sin\left(\pi - \frac{2\pi}{5}\right) \\ &= \sin^{-1} \sin \frac{2\pi}{5} = \frac{2\pi}{5} \end{aligned}$$

Sol. 14. (a)

$$\begin{aligned} \text{LHS} &= \sin^{-1} \frac{4}{5} + \sin^{-1} \frac{3}{5} \\ &= \sin^{-1} \frac{4}{5} + \cos^{-1} \frac{4}{5} = \frac{\pi}{2} \end{aligned}$$

Sol. 15. (c)

$$\tan^{-1}(1+x) + \tan^{-1}(1-x) = \frac{\pi}{2}$$

$$\tan^{-1}\left(\frac{1+x+1-x}{1-(1+x)(1-x)}\right) = \frac{\pi}{2}$$

$$\left[\because \tan^{-1} A + \tan^{-1} B = \tan^{-1}\left(\frac{A+B}{1-AB}\right) \right]$$

$$\tan^{-1}\left(\frac{2}{1-1+x^2}\right) = \frac{\pi}{2}$$

$$\Rightarrow \frac{2}{x^2} = \tan \frac{\pi}{2}$$

$$\Rightarrow x^2 = 0$$

$$\Rightarrow x = 0$$

Sol. 16. (b)

$$x = 4 \tan^{-1}\left(\frac{1}{5}\right) = 2.2 \tan^{-1}\left(\frac{1}{5}\right)$$

$$= 2 \cdot \tan^{-1}\left(\frac{2/5}{1-(2/5)^2}\right)$$

$$= 2 \tan^{-1}(5/12)$$

$$= \tan^{-1}\left(\frac{10/12}{1-\frac{25}{144}}\right)$$

$$= \tan^{-1}\left(\frac{10/12}{119/144}\right) = \tan^{-1}\left(\frac{144/10}{12/119}\right)$$

$$= \tan^{-1}\left(\frac{120}{119}\right)$$

Sol. 17. (c)

$$\tan^{-1}(x-y) = \tan^{-1}\left(\frac{120}{119}\right) - \tan^{-1}\left(\frac{1}{70}\right)$$

$$\tan^{-1}(x-y) = \tan^{-1}\left(\frac{120 - \frac{1}{70}}{1 + \frac{120}{119} \times \frac{1}{70}}\right)$$

$$\begin{aligned} &= \tan^{-1}\left(\frac{120 - \frac{1}{70}}{1 + \frac{120}{119} \times \frac{1}{70}}\right) \\ &= \tan^{-1}\left(\frac{120 \times 70 - 119}{119 \times 70 + 120}\right) = \tan^{-1}\left(\frac{8281}{8450}\right) \end{aligned}$$

Sol. 18. (d)

$$x - y + z = \tan^{-1}\left(\frac{8281}{8450}\right) + \tan^{-1}\left(\frac{1}{99}\right)$$

$$= \tan^{-1}\left(\frac{\frac{8281}{8450} + \frac{1}{99}}{1 - \frac{8281 \times 1}{8450 \times 99}}\right)$$

$$= \tan^{-1}\left(\frac{99 \times 8281 + 8450}{8450 \times 99 - 8281}\right) = \tan^{-1}\left(\frac{828269}{828269}\right)$$

$$= \tan^{-1} 1 = \frac{\pi}{4}$$

Sol. 19. (a)

$$\Rightarrow \tan\left(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4}\right) \dots (i)$$

$$\therefore \frac{1}{5} = \tan^{-1}\left(\frac{2 \times \frac{1}{5}}{1 - \left(\frac{1}{5}\right)^2}\right)$$

$$\left[\because \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} \text{ for } |x| < 1 \right]$$

$$= \tan^{-1}\left(\frac{5}{12}\right)$$

Now, from Eq. (i),

$$\tan\left(\tan^{-1} \frac{5}{12} - \frac{\pi}{4}\right)$$

$$\left[\because \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B} \right]$$

$$\begin{aligned} &= \frac{\tan\left(\tan^{-1} \frac{5}{12}\right) - \tan \frac{\pi}{4}}{1 + \tan\left(\tan^{-1} \frac{5}{12}\right) \cdot \tan \frac{\pi}{4}} \\ &= \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} = \frac{-7}{17} \end{aligned}$$

$$= \frac{5}{12} - 1 = \frac{-7}{17}$$

Sol. 20. (a)

$$1. \text{ LHS} = \sin^{-1} \frac{4}{5} + \sin^{-1} \frac{3}{5}$$

$$= \sin^{-1} \frac{4}{5} + \cos^{-1} \frac{4}{5} = \frac{\pi}{2} = \text{RHS}$$

Statement 1 is correct

2. LHS

$$= \tan^{-1} \sqrt{3} + \tan^{-1} 1 = 60^\circ + 45^\circ = 105^\circ$$

Statement 2 is incorrect

Sol. 21. (d)

We know that,

$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

$$\therefore \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$$

$\therefore$ Statement 1 is incorrect.

Let $x = y$

$$\text{Given that, } \sin^{-1} x + \cos^{-1} y = \frac{\pi}{2} \text{ This is}$$

true for $x \in [-1, 1]$. $\therefore$ Statement 2 is also incorrect.

Sol. 22. (b)

$$\tan^{-1}(\tan \theta) = \theta, -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$\therefore$ Statement 1 is incorrect.

$$\sin^{-1}\left(\frac{1}{3}\right) - \sin^{-1}\left(\frac{1}{5}\right)$$

$$= \sin^{-1}\left[\frac{1}{3} \sqrt{1 - \left(\frac{1}{5}\right)^2} - \frac{1}{5} \sqrt{1 - \left(\frac{1}{3}\right)^2}\right]$$

$$= \sin^{-1}\left[\frac{1}{3} \times \sqrt{\frac{24}{25}} - \frac{1}{5} \sqrt{\frac{8}{9}}\right]$$

$$= \sin^{-1}\left[\frac{1}{3} \times \frac{\sqrt{24}}{5} - \frac{1}{5} \times \frac{\sqrt{8}}{3}\right]$$

$$= \sin^{-1}\left[\frac{\sqrt{24} - \sqrt{8}}{15}\right] = \sin^{-1}\left[\frac{\sqrt{8}(\sqrt{3} - 1)}{15}\right]$$

$$= \sin^{-1}\left[\frac{2\sqrt{2}(\sqrt{3} - 1)}{15}\right]$$

$\therefore$ Statement 2 is correct.

Sol. 23. (d)

$$\cos(2\cos^{-1}(0.8)) = \cos(\cos^{-1}[2(16/25) - 1]) = 0.28$$

Sol. 24. (a)

x, y, z are in GP

$$\text{so } y^2 = xz \dots (i)$$

$\tan^{-1} x, \tan^{-1} y, \tan^{-1} z$ are in AP

$$2 \tan^{-1} y = \tan^{-1} x + \tan^{-1} z$$

$$\tan^{-1}\left(\frac{2y}{1-y^2}\right) = \tan^{-1}\left(\frac{x+z}{1-xz}\right)$$

$$\frac{2y}{1-y^2} = \frac{x+z}{1-xz} \dots (ii)$$

By (i) and (ii)

$$\frac{2y}{1-xz} = \frac{x+z}{1-xz}$$

$$2y = x+z$$

So x, y, z are in AP

We conclude that x, y, z are in both AP and GP

It is only possible when $x = y = z$

Sol. 25. (b)

$$\Rightarrow \sin^{-1} \frac{3}{5} + \tan^{-1} \frac{1}{7}$$

$$\Rightarrow \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{1}{7} = \tan^{-1}\left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \cdot \frac{1}{7}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \cdot \frac{1}{7}}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Sol. 26. (b)

$$\text{Range of } \sin^{-1} x \text{ is } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Sol. 27. (c)

$$\sin^{-1} \sin \left(\frac{2\pi}{3} \right) = \sin^{-1} \sin \left(\pi - \frac{\pi}{3} \right)$$

$$\sin^{-1} \sin \left(\frac{\pi}{3} \right) = \frac{\pi}{3}$$

Sol. 28. (b)

$$\Rightarrow \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5}$$

$$\Rightarrow \tan^{-1} \left(\frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \cdot \frac{3}{5}} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Sol. 29. (a)

$$\tan^{-1} \left(\frac{2x+3x}{1-2x \cdot 3x} \right) = \frac{\pi}{4}$$

$$\frac{5x}{1-6x^2} = 1$$

by solving this quadratic equation

$x = 1/6$ and -1 [1 rejected]

Sol. 30. (d)

$$\sin^{-1} \frac{4}{5} + \cos^{-1} \frac{4}{5} = \frac{\pi}{2}$$

$$\frac{\pi}{2} - \frac{\pi}{2} = 0$$

Sol. 31. (b)

$$2 \tan^{-1} p - 2 \tan^{-1} q = 2 \tan^{-1} x$$

$$2 \tan^{-1} \left(\frac{p-q}{1+pq} \right) = 2 \tan^{-1} x$$

$$\left(\frac{p-q}{1+pq} \right) = x$$

Sol. 32. (b)

$$\tan \left\{ 2 \tan^{-1} \frac{1}{3} \right\} = \tan \left(\tan^{-1} \frac{\frac{2}{3}}{1 - \frac{1}{9}} \right)$$

$$\frac{2/3}{8/9} = 3/4$$

Sol. 33. (b)

$$\sin^{-1} x - \cos^{-1} x = \pi/6 \dots (1)$$

we know that

$$\sin^{-1} x + \cos^{-1} x = \pi/2 \dots (2)$$

by solving above two equations we get

$$x = \pi/3 \text{ (unique solutions)}$$

Sol. 34. (a)

Sol. 35. (a)

$$\cot \left[\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right]$$

$$= \cot \left[\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} \right]$$

$$= \cot \left[\tan^{-1} \left(\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}} \right) \right]$$

$$= \cot \left[\tan^{-1} \left(\frac{17}{6} \right) \right] = \cot \left[\cot^{-1} \left(\frac{6}{17} \right) \right] = \frac{6}{17}$$

Sol. 36. (c)

$$\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$$

$$\sin^{-1} x = \sin^{-1} y = \sin^{-1} z = \pi/2$$

$$x = y = z = 1$$

$$n^{1000} + y^{1001} + z^{1002} = 1 + 1 + 1 = 3$$

Sol. 37. (a)

$$\tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{x}{3} \right) = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{\frac{1}{2} + \frac{x}{3}}{1 - \frac{1}{2} \cdot \frac{x}{3}} \right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{\frac{3+2x}{6-x}}{\frac{6-x}{6}} = 1 \Rightarrow 3+2x = 6-x$$

$$3x = 3$$

$$x = 1$$

Sol. 38. (c)

$$3 \sin^{-1} x + \cos^{-1} x = \pi$$

$$2 \sin^{-1} x + \sin^{-1} x + \cos^{-1} x = \pi$$

$$2 \sin^{-1} x + \frac{\pi}{2} = \pi$$

$$2 \sin^{-1} x = \frac{\pi}{2} \Rightarrow \sin^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow x = \frac{1}{\sqrt{2}}$$

Sol. 39. (d)

$$\tan^{-1} \cot (\operatorname{cosec}^{-1} 2)$$

$$\tan^{-1} \cot (30^\circ)$$

$$\tan^{-1} \tan (90^\circ - 30^\circ) = 60^\circ$$

Sol. 40. (c)

$$2 \cot \left(\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3} \right)$$

$$\text{Let } \Rightarrow \left(\cos^{-1} \frac{\sqrt{5}}{3} \right) = \theta$$

$$2 \cot \left(\frac{\theta}{2} \right) = 2 \sqrt{\frac{1+\cos \theta}{1-\cos \theta}}$$

$$2 \sqrt{\frac{1+\cos \left(\cos^{-1} \frac{\sqrt{5}}{3} \right)}{1-\cos \left(\cos^{-1} \frac{\sqrt{5}}{3} \right)}} = 2 \sqrt{\frac{1+\frac{\sqrt{5}}{3}}{1-\frac{\sqrt{5}}{3}}}$$

$$= 2 \sqrt{\frac{1+\frac{\sqrt{5}}{3}}{1-\frac{\sqrt{5}}{3}}} = 2 \sqrt{\frac{3+\sqrt{5}}{3-\sqrt{5}}}$$

$$= 2 \sqrt{\frac{3+\sqrt{5}}{3-\sqrt{5}}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}} = 2 \times \frac{3+\sqrt{5}}{2} = 3+\sqrt{5}$$

Sol. 41. (a)

$$\sec^{-1} p - \operatorname{cosec}^{-1} q = 0$$

$$\sec^{-1} p = \operatorname{cosec}^{-1} q$$

we know that $\sec x$ and $\operatorname{cosec} x$ are equal at 45°

$$\text{so } p = q = \frac{1}{\sqrt{2}}$$

$$p^2 = q^2 = \frac{1}{2} + \frac{1}{2} = 1$$

Sol. 42. (a)

$$1 + \sin^2 \left(\cos^{-1} \left(\frac{3}{\sqrt{17}} \right) \right) = 1 + \sin^2 \left(\cos^{-1} \left(\frac{3}{\sqrt{17}} \right) \right)$$

$$= 1 + \sin^2 \left(\sin^{-1} \left(\frac{\sqrt{8}}{\sqrt{17}} \right) \right) = 1 + \left(\frac{\sqrt{8}}{\sqrt{17}} \right)^2$$

$$= 1 + \frac{8}{17} = \frac{25}{17}$$

Sol. 43. (c)

$$\sqrt{15 + \cot^2 \left(\frac{\pi}{4} - 2 \cot^{-1} 3 \right)}$$

$$\sqrt{15 + \cot^2 \left(\frac{\pi}{4} - 2 \tan^{-1} \frac{1}{3} \right)}$$

$$\sqrt{15 + \cot^2 \left(\tan^{-1} 1 - \tan^{-1} \frac{3}{1 - \frac{1}{9}} \right)}$$

$$\sqrt{15 + \cot^2 \left(\tan^{-1} 1 - \tan^{-1} \frac{3}{4} \right)}$$

$$\sqrt{15 + \cot^2 \left(\tan^{-1} \left(\frac{1 - \frac{3}{4}}{1 + \frac{3}{4}} \right) \right)}$$

$$\sqrt{15 + \cot^2 \left(\tan^{-1} \left(\frac{1}{7} \right) \right)}$$

$$\sqrt{15 + \cot^2 (\cot^{-1} 7)}$$

$$\sqrt{15 + 49} = 8$$

Sol. 44. (b)

$$\tan^{-1} \left(\frac{a}{b} \right) - \tan^{-1} \left(\frac{a-b}{a+b} \right)$$

$$\Rightarrow \tan^{-1} \left(\frac{\frac{a}{b} - \frac{a-b}{a+b}}{1 + \left(\frac{a}{b} \right) \left(\frac{a-b}{a+b} \right)} \right)$$

$$\Rightarrow \tan^{-1} \left(\frac{a(a+b) - b(a-b)}{b(a+b) + a(a-b)} \right)$$

$$\Rightarrow \tan^{-1} \left(\frac{a^2 + b^2}{a^2 + b^2} \right) = \tan^{-1} 1 = \frac{\pi}{4}$$

Sol. 45. (c)

$$\sin^{-1} x = \cos^{-1} x$$

$$\Rightarrow \sin^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

$$\Rightarrow 2 \sin^{-1} x = \frac{\pi}{2} \Rightarrow \sin^{-1} x = \frac{\pi}{4}$$

$$\Rightarrow x = \frac{1}{\sqrt{2}}$$

Sol. 46. (c)

$$4 \sin^{-1} x + \cos^{-1} x = \pi \dots \dots (i)$$

we know that

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \dots \dots (ii)$$

eq(i) - eq(ii)

$$3 \sin^{-1} x = \frac{\pi}{2}$$

$$\sin^{-1} x = \frac{\pi}{6} \Rightarrow x = \frac{1}{2}$$

$$\sin^{-1} x + 4 \cos^{-1} x = \frac{\pi}{6} + 4 \left(\frac{\pi}{3} \right) = \frac{3\pi}{2}$$

Sol. 47. (b)

$$\cot^2(\sec^{-1} 2) + \tan^2(\operatorname{cosec}^{-1} 3)$$

by Pythagoras theorem

$$\cot^2 \left(\cot^{-1} \left(\frac{1}{\sqrt{3}} \right) \right) + \tan^2 \left(\tan^{-1} \left(\frac{1}{\sqrt{8}} \right) \right)$$

$$= \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{8}} \right)^2 = \frac{1}{3} + \frac{1}{8} = \frac{11}{24}$$

Sol. 48. (d)

$$\tan^{-1} k + \tan^{-1} \left(\frac{1}{k} \right)$$

$$\tan^{-1} k + \cot^{-1} k = \frac{\pi}{2}$$

Sol. 49. (c)

$$\tan^{-1} k + \tan^{-1} \left(\frac{1}{2} \right) = \tan^{-1} \left(\frac{k + \frac{1}{2}}{1 - \frac{k}{2}} \right) = \frac{\pi}{4}$$

$$\Rightarrow \left(\frac{k + \frac{1}{2}}{1 - \frac{k}{2}} \right) = 1 \Rightarrow k + \frac{1}{2} = 1 - \frac{k}{2} \Rightarrow k = \frac{1}{3}$$

Sol. 50. (a)

$$\cot^{-1} 9 + \operatorname{cosec}^{-1} \left(\frac{\sqrt{41}}{4} \right)$$

$$\Rightarrow \tan^{-1} \left(\frac{1}{9} \right) + \tan^{-1} \left(\frac{4}{5} \right)$$

$$\Rightarrow \tan^{-1} \left(\frac{\frac{1}{9} + \frac{4}{5}}{1 - \frac{1}{9} \times \frac{4}{5}} \right) = \tan^{-1} \left(\frac{41}{41} \right)$$

$$\Rightarrow \tan^{-1}(1) = \frac{\pi}{4}$$

Sol. 51. (a)

$$\tan \left[\frac{1}{2} \sec^{-1} \left(\frac{2}{\sqrt{3}} \right) \right] = \tan \left[\frac{1}{2} \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) \right]$$

$$\tan \left[\frac{1}{2} \cdot \frac{\pi}{6} \right] = \tan \left(\frac{\pi}{12} \right) = 2 - \sqrt{3}$$

Video solutions

[Inverse trigonometry 2011 to 2022 questions](#)

[Inverse trigonometry 2023 to 2024 questions](#)